



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000
WEEK 4.09

TUTOR GUIDE

CASE ONE

Short case number: 4_09_01

Category: Clinical Skills

Discipline: Emergency Medicine

Setting: Emergency Department_urban

Topic: Approach to the patient with Decreased Level of Consciousness

CASE

A 43 year old male is brought into the emergency department with a decreased level of consciousness. The ambulance officers have no information regarding his past history. He was found in a park by passers-by. He is confused and unable to answer questions.

Observations on Arrival:

GCS is 13 (E3V4M6)

Pulse rate 92 bpm

BP 110/70

Examination reveals bruising around the chest wall.

His abdomen is soft and there are no obvious injuries in the upper or lower limbs.

QUESTIONS

1. Outline your approach to the patient with decreased level of consciousness. What are some of the potential causes of depressed level of consciousness in this patient?
2. Outline the investigations that you may need to perform, and justify the ordering of these investigations
3. Outline your initial management of this patient, whilst you await investigation results.

FURTHER CASE INFORMATION

Results of Blood Investigations:

Sodium 114mmol/L (135-145)

Potassium 3.3 mmol/L (3.5-5.0)

Chloride 101mmol/L (97-109)

Urea 6.1mmol/L (3-8)

Creatinine 85 micromol/L (70-110)

QUESTIONS

4. Your blood results return. All results are normal except for the electrolytes, which are shown above describe the results. Could this be contributing to his decreased level of consciousness outline your thoughts on the cause of his altered conscious state.

ANSWERS

Outline your approach to the patient with decreased level of consciousness. What are some of the potential causes of depressed level of consciousness in this patient?

There are many potential differentials here and can be approached as intracranial / extracranial.

Intracranial causes include:

- trauma and associated bleeds
- tumours, that may be directly affecting conscious state or causing a seizure
- Infective causes- such as Meningitis and Encephalitis
- There may be trauma involved without obvious signs on CT such as diffuse axonal injury

Extracranial Causes include:

- Cardiac – including hypotension, arrhythmias
- Respiratory – hypoxia, hypercapnoea
- Toxic – intoxication with alcohol, drugs such as sedatives, narcotics,
- Metabolic/Endocrine –
 - Failure –hepatic/renal
 - Hyper/hypo -
 - glycaemia
 - natraemia
 - calcaemia
 - thyroid
- Psychiatric

Consider a combination of issues, including if the patient has had a seizure, secondary to some other cause.

Outline the investigations that you may need to perform, and justify the ordering of these investigations

Lab tests

- Full blood count – consider infection, or low platelet count that may be associated with intracranial bleed
- Electrolytes – To look at the many possible causes of depressed conscious state, such as hyponatraemia, renal failure etc.
- Liver Function Tests – consider liver failure, encephalopathy
- Thyroid Function tests
- Coagulation – to look any hyper or hypocoag states that may lead to thrombosis or bleed.
- Blood sugar – the patient may be hypoglycaemic, or hyperglycaemic ie., in DKA or HONKC

- Other tests may need to be done, in association with history, other results and consultation. These may include HIV testing, syphilis serology etc.
- Lumbar Puncture may be required in this patient.
- If the patient is found to be on antiepileptic medications, then levels.

Radiology

- Chest xray – looking for pneumonias and other causes that may be leading to hypoxia.
- CT Head, looking for a bleed, or a tumour.

Outline your initial management of this patient, whilst you await investigation results.

Initial ABC, to ensure control of life threatening conditions.

Consider this to be similar to a trauma case and do a full primary and secondary survey.

Establish as much history as possible and do a bedside blood sugar level, or ask if the ambulance has done one. If the patient is hypoglycaemic administer 50mL of 50% glucose

Has the conscious state changed since found?

Check the patient for any evidence of past medical history eg alert bands or something in the wallet.

Consider a cervical collar as there has obviously been trauma to the chest.

Examine the patient, especially looking at airway and breathing and circulation, ensure appropriate heart rate and blood pressure and place patient on monitor looking for an arrhythmia.

Look externally for any needle track marks.

Look for eye signs – pupil size – dilated or pinpoint

Investigations to consider are:

FBE, EUC, BSL, LFT's, CXR, ECG, EEG, CT head

Consider acutely reversing drug effects of narcotics including heroin, with naloxone 400mcg IV

Describe your results. Could this be contributing to his decreased level of consciousness your thoughts on the cause of his altered conscious state.

Bloods show a hyponatraemia, which can explain all the symptoms. The potential causes of this must be considered including primary polydipsia and diabetes insipidus. Correction of hyponatraemia must be in consultation with nephrology and care should be taken to avoid rapid correction.

CASE TWO

Short case number: 4_09_02

Category: Clinical Skills

Discipline: Emergency Medicine

Setting: Emergency Department

Topic: Head injury

CASE

A 32 year old intoxicated male is brought in by ambulance following a 10 metre fall. The ambulance officers state he had been on the balcony drinking, and was witnessed to fall over the edge. The landing was on a concrete path below.

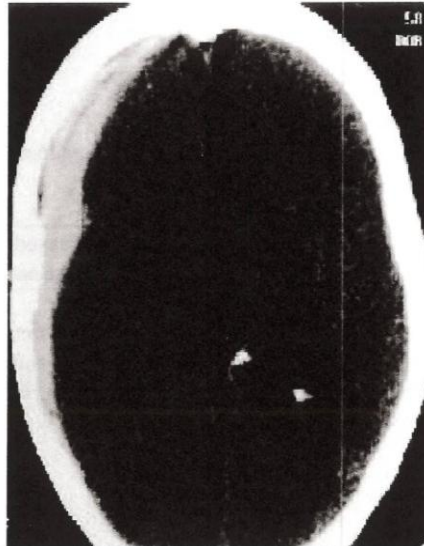
Observations on arrival are: Heart rate of 105bpm, BP 120/75, Oxygen Saturations 97% on room air. GCS 12/15. You note on arrival that he has a large scalp laceration.

QUESTIONS

1. Outline your initial assessment and management of this patient.
2. Outline the concepts of primary and secondary brain injury secondary to trauma.
3. Outline the possible causes of decreased GCS in the setting of traumatic head injury.

CASE

You perform a Head CT which is as follows.



QUESTIONS

- | |
|---|
| <p>4. Briefly describe the above CT findings.</p> <p>5. What are the treatment options for the above diagnosis?</p> |
|---|

CASE

The patient returns to the emergency department following the CT scan. A short while later, the nurse calls for assistance as the patient is now totally unresponsive with a GCS of 3. She notes that he also has a dilated, unresponsive right pupil.

QUESTIONS

- | |
|--|
| <p>6. What immediate management is required in this patient?</p> |
|--|

ANSWERS

1. Outline your initial assessment and management of this patient.

This is a case of a potential major trauma. The head injury is certainly a major component, however beware not to miss other injuries.

This patient will require a full Primary and Secondary Survey.

Initial ABC, with cervical spine precautions. Must ensure airway patent and protected and C-spine is protected. Also check circulation and as part of that look at the severity of the scalp wound as there may be significant blood loss through this. Put a compression bandage on this, suture or staple. Complete Primary survey and ensure no other injuries on log roll. Perform a FAST examination and initial trauma series Xrays.

Obtain a minimum of one large bore cannulae in the patient and take bloods including a bedside BSL. Also check for pupillary reaction and perform a neurological examination.

The patient will need a CT head and cervical spine, however will initially need a chest and pelvis Xray in the resus cubicle.

2. Outline the concepts of primary and secondary brain injury secondary to trauma.

Primary Brain injury occurs secondary to the direct effects of the trauma, be they haematoma or contusion, or other direct brain damage.

Secondary Brain injury occurs due to hypoxia or hypoperfusion

3. Outline the possible causes of decreased GCS in the setting of traumatic head injury.

Loss of consciousness may or may not occur in head injury and the length of this depressed GCS may vary.

A loss of consciousness occurs due to the interruption of the regular brain's function. This can occur secondary to a brief violent movement of the brain within the cranial vault.

Injury to parts of the brain due to direct damage such as a bleed into the brain matter can also affect conscious state.

Depressed conscious state can occur when there is a bleed displacing the brain such as with a subdural. Remember the cranial vault has a fixed volume.

Depressed conscious state can also occur due to brain damage causing swelling and increased intracranial pressure, to a point where there is minimal brain perfusion. Further, depressed conscious state can occur when there is a bleed within the brain that causes any pressure on the ventricles (especially the 4th ventricle) causing a hydrocephalus.

4. Briefly describe the above CT findings.

The CT head shows an acute subdural haemorrhage. Subdural diagnosis is made primarily due to the shape and the distribution of the haemorrhage- it is concave and associated with the brain parenchyma (as compared to extradural which is convex and usually has a dural separation from the brain parenchyma). This is secondary to tearing of veins that drain blood to dural sinuses. There is occasionally an arterial bleed involved resulting in significant and rapid bleeding. Subdurals are associated with a high mortality rate – up to 70% and need early evacuation.

5. What are the treatment options for the above diagnosis?

The patient will need operative treatment with craniotomy and drainage. There is an urgency for this to occur as there is a depressed level of consciousness and a midline shift in the brain. There may also be a need for an osmotic agent such as mannitol. Intravenous mannitol will allow reduction of fluid in the brain and decreased intracranial pressure.

There may also be a need to intubate and paralyse the patient as this also decreases the intracranial pressure.

6. What immediate management is required in this patient?

The patient's GCS has dropped which indicates a potential increased in bleeding, to the point that it impairs adequate brain function. Ensure that there is no other cause such as hypotension, hypoglycaemia etc.. The GCS is now <9 which indicates that the airway is not protected. Rapid sequence intubation will be required, with in-line immobilisation, to protect the cervical spine. Beware to not cause hypotension with the medications given to ensure that there is no secondary hypoxic brain injury. The patient then needs to be paralysed and run normocarbic with a potential for slight head-up on the bed. The definitive treatment is evacuation, however discussion about the use of mannitol must be made with the neurosurgical consultant.

CASE THREE

Short case number: 4_09_03

Category: Clinical Skills

Discipline: Emergency Medicine

Setting: Emergency Department

Topic: Spinal Cord Injury

CASE

A 27 year old male is brought to the emergency department following an assault. He has been hit to the head and neck with a metal bar.

On arrival, his GCS is 14, pulse rate is 96bpm, BP is 135/70 and saturations are 100% on room air. He is complaining of neck pain and weakness in his arms.

QUESTIONS

1. Outline your initial concerns, assessment and management regarding this patient in view of the above information.

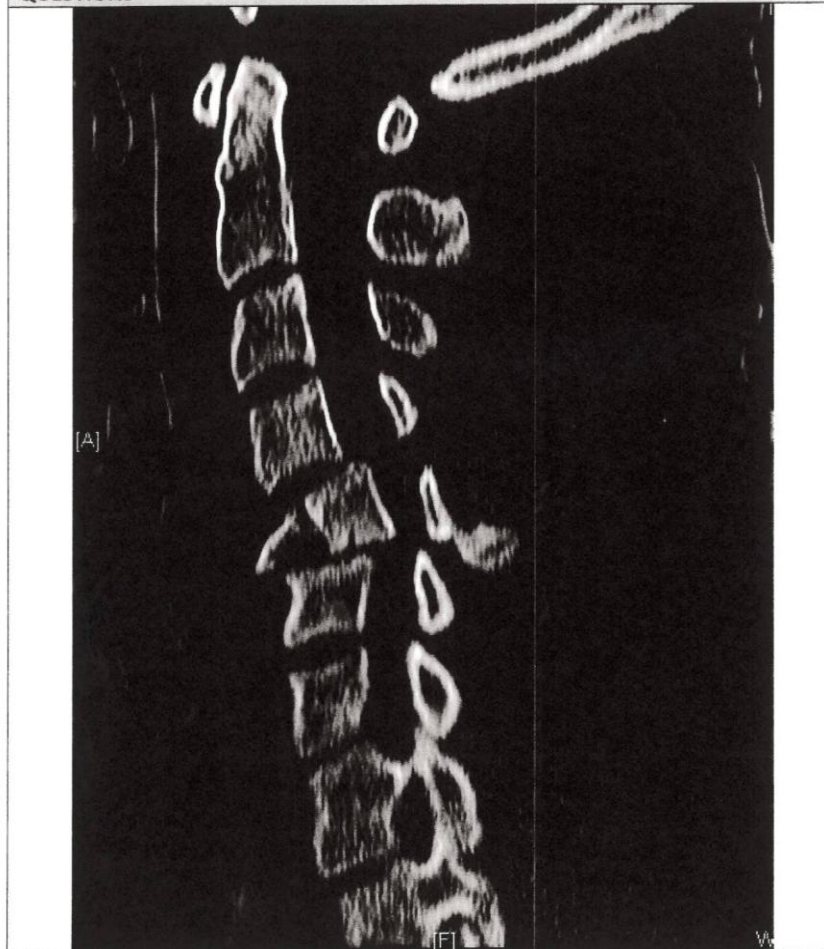
CASE

Your neurological examination of the limbs reveals normal tone bilaterally, but decreased power and some sensory abnormality over both shoulders of the C5/6 dermatomes.

QUESTIONS

2. What investigations may be necessary given the findings?
3. A CT slice is shown below. Please comment on the injury.
4. What is spinal shock? Is this patient at risk for developing this condition?
5. Outline the appropriate disposition of this patient.

QUESTIONS



ANSWERS

1. Outline your initial concerns, assessment and management regarding this patient in view of the above information.

This is a trauma patient and should be approached with a full primary and secondary survey, with particular attention to spinal precautions. The main concern with this patient is a spinal cord injury.

Large bore cannulae

FAST ultrasound

Initially we need to assess the patient as we would any other trauma patient looking at the ABC and ensuring cervical spine precautions.

Careful neurological examination is required to document exactly what is needed in terms of investigations.

2. What investigations may be necessary given the findings?

The patient will need a trauma series as guided by the findings on examination. Certainly consideration should be given to a chest Xray as a minimum. The patient will require a CT head and CT cervical spine, to assess immediate fractures, however given clinical findings, an MRI may be necessary. Beware that if there is one fracture then there is a 10% chance of second fracture somewhere else in the spine. It would be appropriate to either CT or Xray the rest of the spine, as appropriate with the examination.

3. A CT slice is shown below. Please comment on the injury.

There is a significant fracture and displacement of C5. The anterior border of C5 is fractured and displaced and there is a posterior displacement and potential impingement of the spinal cord by the vertebral body.

4. What is spinal shock and Neurogenic shock? Is this patient at risk for developing this condition?

Spinal shock is the condition where by the patient has motor, sensory and reflexes loss due to cord damage. The reflexes gradually return.

Yes the patient is at risk.

Neurogenic shock results from autonomic dysfunction. Secondary to this there is hypotension, bradycardia, peripheral vasodilatation and hypothermia. It usually occurs with spinal cord injuries above T6

5. Outline the appropriate disposition of this patient.

Consultation with neurosurgery early.

Transfer to a dedicated Spinal Unit

Check three measurements in C-spines:
3mm max between dens and C2
7mm soft tissue anterior to high vertebra
21mm in front of vertebra lower C-spine

CASE FOUR

Short case number: 4_09_04

Category: Clinical Skills

Discipline: Emergency Medicine

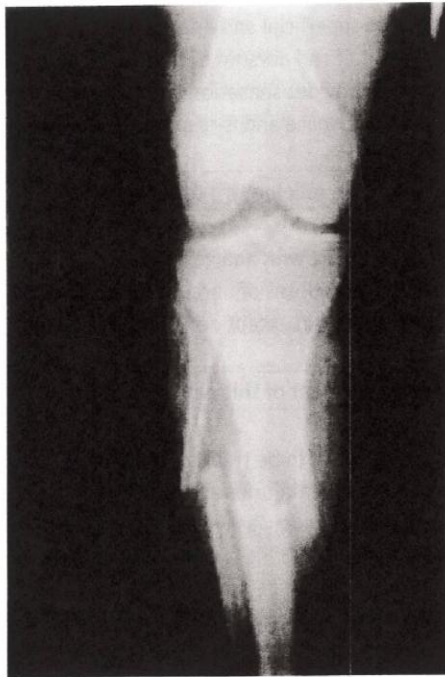
Setting: Emergency Department

Topic: Peripheral Nerve Injury

CASE

A 19 year old football player presents following being tackled in a rugby match. He has a painful lateral knee and has developed a foot-drop on that side. He has no other injuries.

Radiograph of the Knee is as follows:



QUESTIONS

1. You are asked to call the Orthopaedic Registrar with the Radiograph findings. Describe the findings as you would over the telephone to the Registrar. What is the injury that has caused the footdrop? Outline the clinical signs you would look for.
2. Outline your approach to pain relief in this patient.
3. What is your general management of this patient?

ANSWERS

1. You are asked to call the Orthopaedic Registrar with the Radiograph findings. Describe the findings as you would over the telephone to the Registrar.

There is a proximal fibula fracture with associated tibial fracture. There is overriding of the bone fragments. A further xray at right angles will be required to determine true angulation.

2. What is the injury that has caused the footdrop? Outline the clinical signs you would you look for.

The footdrop is most likely caused by injury to the common peroneal nerve. The common peroneal nerve is one of the branches of the sciatic nerve, the other is the tibial nerve. The common peroneal nerve passes into the peroneous longus muscle and there divides into superficial and deep branches. The superficial branch provides sensation to lateral calf and dorsum of foot and is responsible for eversion of the foot. The deep branch provides sensation to a small area of web space between the 1st and 2nd toe webspace and is responsible for dorsiflexion of the foot.

3. Outline your approach to pain relief in this patient.

There are various methods to assist with analgesia. Firstly splint the leg using a plaster of paris. Intravenous bolus doses of narcotic as needed for pain relief. No other methods will assist in this case, except definitive treatment.

4. What is your general management of this patient.

This is a significant injury and needs to be treated as a trauma, with primary and secondary survey to ensure no other injuries are missed. ABC and cervical spine precautions apply as well as FAST ultrasound and a trauma series.

Intravenous access must be established with a minimum of one large bore cannula. Examine the limb to ensure it is neurovascularly intact and following adequate analgesia, perform a neurological examination. Ensure there is no early compartment syndrome and consult with orthopaedic team as this patient will require operative treatment.